

Documentation (Roman Urdu)

Yeh file pure application ki **Configuration Center** (Settings) ki tarha kaam karti hai. Project ki jitni bhi basic configuration variables hain (directories ke rastay, models ke naam, booleans/switches) wo isi ek file mein manage hotay hain.

File Ka Maqsad (Purpose)

Is project mein hardcoded values likhne (misal ke t■■■ per url="http://localhost:11434") ke bajaye Pydantic ki `BaseSettings` class ka istemal kiya gaya hai. Iska maqsad yeh hai ke: 1. Saari global settings ek hi jagah mojud hon. 2. Agar user `.env` file (`SOLLOLM_` prefix ke sath) banaye, toh wo idher automatically inject ho jaye aur settings ko override kar de. 3. Type safety rahay, i.e., integers properly validate hon jisme string values na asakein.

Code Blocks aur Functions ki Tafseel

Is file mein bas ek hi body / class mojud hai, jiske variable sections ko yahan tafseel se bayan kiya ja raha hai.

1. class Settings(BaseSettings)

Kya karta hai?: Yeh pydantic module se inherit honi wali master object class hai jo app configurations store karti hai:

- **Application & Paths:** Project root folder `base_dir`, database ka rasta (`data/db/solollm.db`) batati hai.
- **Ollama Configurations:** Default local LLM kounsa hoga (`llama3.2:1b`), Ollama kahan background mein store hoga (`data/ollama`), auto-start option.

Inference (Generation) Settings:

- `max_tokens=2048` aur `temperature=0.7` AI ke text flow control karte hain.
- `gpu_layers` (Model GPU memory mein jayega y nai) aur `max_power_mode` control karte hain.
- **Continuation:** Agar response kaat (truncate) jaye, toh overlapping chars kitne hongy usay auto-continue karne k lite.
- **Context Distillation (Phase 3):** LLM inputs/prompts ko shrink aur compressed rakhne ke parameters (Chain of density, deduplication, thresholds).

- **Knowledge Graph (Phase 4):** Web scraping limits (MB timeout) aur graph retrieval switches.
- **RAG Engine Constants:** Defaults set karta hai ke kaunsa vector embedding model download hoga (all-MiniLM-L6-v2), aur faiss backend chalega. Chunk sizes kitni baray hongy.
- **Agent Framework (Phase 5):** Agents kis temprature pe deal kareingy aur inke default tools (calculator, rag_search, web_scrapping) console defined rehtey hain.

2. File execution script `settings = Settings()`

- **Kya karta hai?:** Class ka instance object (settings) create karta hai jo baad mein poori app me import (from core.config import settings) kar k use kia jata ha.
- **Path directories initialize karana:** Jab app pehli baar chalegi, aur agar data/ aur uske andar k folders (db, documents, cache, ollama) nae bne huway toh `.mkdir(parents=True, exist_ok=True)` commands k through yeh un directories/folders ko storage ke andar physically create kar deta hai taakay app code crash na ho file not found aane pe.